

CASE STUDY:

Harnessing Wind Energy At Tareybhiri: A Success Story



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Tareybhiri is a tourist attraction in south Sikkim. It has a clear view of wind path with no obstruction by trees and nearby mountains, and the wind is generally accounted to be high throughout the year. Wind-energy harnessing at Tareybhiri with solar redundancy has been done to bring awareness about higher-capacities wind turbine system in this area.

Usually local wind prevails in the area due to pressure differences between plains and mountains. Quite a number of wind-speed data studies were carried out by some agencies and individuals, but further initiatives could not be initiated. The main hindrance was a huge investment and the risk of failure of the project.

After site inspection, I thought about how this wind power could be tapped for electricity generation. So, I started surveying and studying the site and its feasibility

for wind turbine installation. I contacted everyone who had studied the area for wind power, but could not access the wind-speed data.

The first bit of information needed for wind turbine installation is the wind-speed data for at least one year. For this, I took my digital anemometer to the spot where I wished to install my turbine. With this preliminary data, I conducted a feasibility-analysis study using some basic methodology of statistics and probability density function, which concluded my approximated results as follows:

1. The average wind speed was 6m/s
2. The average power in the wind was 113W/m²
3. The annual average energy generated was 1800kWh/year for wind turbines of 0.9kW

The next step was to get a turbine, its accessories, and get it installed. I needed funds for the project and no organisation was willing to take on the risk. Finally, I met with Siphora G. Targain, SREDA director, at Gangtok, Sikkim, on September 16, 2013. She gave me access to the ten-year-old, dead, unutilised wind-solar hybrid system at Assangthang (another village in south Sikkim), which was installed in 2004 for CFL street-light lighting purpose. It ran successfully for around three years and then went dead and was lying unutilised. The land owner, along with the local *panchayat*, wanted it removed from the site. This is when I decided to repair this damaged system.

The next initiative required for the project was manpower, and the small budget was just sufficient enough to complete the task. For this, I gave presentations, took part in debates and tried to inspire my fellow workers, colleagues and friends for social benefit, self-interest and hands-on experience involved in the project. This way, I gathered a team of 30 people who con-

Fig. 1: Wind turbine with solar redundancy system

